

ASSIGNMENT 3

Name: Manvendra Singh Tomar

Roll No.: 241500188

Q1 — Question

Task 1 – Project Workspace: Create the project workspace with src, docs, config and backup directories and library_info.txt. Demonstrate pwd, directory navigation, listing files, displaying file contents and directory structure.

```
Manvendra@LAPTOP-PK65VJLN MINGW64 ~
$ mkdir Manvendra-GLA-LibrarySystem

Manvendra@LAPTOP-PK65VJLN MINGW64 ~
$ cd Manvendra-GLA-LibrarySystem

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem
$ mkdir src docs config backup

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem
$ touch library_info.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem
$ nano library_info.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem
$ pwd
/c/Users/Manvendra Singh Tomar/Manvendra-GLA-LibrarySystem

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem
$ ls
backup/  config/  docs/  library_info.txt  src/

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem
$ cat library_info.txt
Name: Manvendra Singh Tomar
Roll Number: 241500188
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem
$ find .
.
./backup
./config
./docs
./library_info.txt
./src
```

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Q2 — Question

Task 2 – Git Repository: Initialize Git, configure username/email, check status, stage files, create the initial commit, modify library_info.txt, add the project objective and description, check changes, stage the modified file, create a second commit and display commit history.

```
Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git init
Initialized empty Git repository in C:/Users/Manvendra/Manvendra-GLA-LibrarySystem/.git/

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git config --global user.name "Manvendra Singh Tomar"

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git config --global user.email "manvendrasinghtomar2005@gmail.com"

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to track)

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git add .
bash: git: command not found

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git add .
warning: in the working copy of 'library_info.txt', LF will be replaced by CRLF the next time Git touches it

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git commit -m "Initial project setup"
[master (root-commit) 22551a7] Initial project setup
 1 file changed, 5 insertions(+)
 create mode 100644 library_info.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ nano library_info.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git diff
diff --git a/library_info.txt b/library_info.txt
index 739e85b..a12ace7 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -1,3 +1,10 @@
+Project Name: GLA Digital Library System
+
+Project Objective:
+To develop an online library management system for managing books and library-related operations efficiently.
+
+Project Description:
+The system will help manage library books and provide basic book management functionality.

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git add library_info.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git commit -m "Add project objective and description"
[master 5b2e714] Add project objective and description
 1 file changed, 7 insertions(+)

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git log --oneline
5b2e714 (HEAD -> master) Add project objective and description
22551a7 Initial project setup

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$
```

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Q2 — Question

Task 2 – Git Repository: Initialize Git, configure username/email, check status, stage files, create the initial commit, modify library_info.txt, add the project objective and description, check changes, stage the modified file, create a second commit and display commit history.

```
Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ cat library_info.txt
Project Name: GLA Digital Library System

Project Objective:
To develop an online library management system for managing books and library-related operations efficiently.

Project Description:
The system will help manage library books and provide basic book management functionality.

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git add library_info.txt
warning: in the working copy of 'library_info.txt', LF will be replaced by CRLF the next time Git touches it

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git commit -m "Add project objective and description"
[master 5b2e714] Add project objective and description
1 file changed, 8 insertions(+)

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git log --online
fatal: unrecognized argument: --online

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ |
```

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Q3 — Question

Task 3 – Feature Branch: Develop a Book Management Module on the book-feature branch. Create src/book.txt with a description and at least three features. Stage and commit the changes and demonstrate branch creation, switching, current branch, all branches and returning to main/master.

```
Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git branch book-feature

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git switch book-feature
Switched to branch 'book-feature'

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (book-feature)
$ git branch
  book-feature
* master

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (book-feature)
$ nano src/book.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (book-feature)
$ git add src/book.txt
warning: in the working copy of 'src/book.txt', LF will be replaced by CRLF the next time Git touches it

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (book-feature)
$ git commit -m "Add Book Management Module"
[book-feature 7a2c8dc] Add Book Management Module
1 file changed, 10 insertions(+)
create mode 100644 src/book.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (book-feature)
$ git branch --show-current
book-feature

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (book-feature)
$ git branch
* book-feature
  master

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (book-feature)
$ git switch main
fatal: invalid reference: main

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (book-feature)
$
```

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Q4 — Question

Task 4 – Merge Conflict: Merge book-feature into main/master, create/use library-update, modify the same line in library_info.txt differently on the two branches, commit both changes, attempt the merge, identify and manually resolve the conflict, then stage, commit and display the final file, status and history.

```
Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git switch master
git merge book-feature
Already on 'master'
Already up to date.

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git switch -c library-update
fatal: a branch named 'library-update' already exists

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ nano library_info.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git add library_info.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git commit -m "Update project name in library"
On branch master
nothing to commit, working tree clean

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git switch master
Already on 'master'

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ nano library_info.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git add library_info.txt

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git commit -m "Modify project name in master"
On branch master
nothing to commit, working tree clean

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git merge library-update
Already up to date.

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ cat library_info.txt
Name: Manvendra Singh Tomar
Roll Number: 241500188
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Semester: 5th

Project Objective:
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```

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Q4 — Question

Task 4 – Merge Conflict: Merge book-feature into main/master, create/use library-update, modify the same line in library_info.txt differently on the two branches, commit both changes, attempt the merge, identify and manually resolve the conflict, then stage, commit and display the final file, status and history.

```
On branch master
nothing to commit, working tree clean

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git merge library-update
Already up to date.

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ cat library_info.txt
Name: Manvendra Singh Tomar
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Subject: Application Build and Release Automation Lab
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Semester: 5th

Project Objective:
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Project Description:
The system will help manage library books and provide basic book management functionality.

Project Name: GLA Digital Library System

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git status
On branch master
nothing to commit, working tree clean

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git log --oneline --graph --all
* c215eec (HEAD -> master) Finalize conflict resolution
* b02a695 Clean final merge resolution
| ac43de8 Resolve merge conflict
| \
| * 4c6466a (library-update) Update project name in library
| * 9704312 Modify project name in master
| /
* 7a2c8dc (book-feature) Add Book Management Module
| 5b2e714 Add project objective and description
| 22551a7 Initial project setup

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$
```

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Q5 — Question

Task 5 – Submission: Submit screenshots of important commands, the project directory, Git history, branch information, merge demonstration, and conflict and resolution demonstration. Students must be able to explain the commands used.

```
Manvendra@LAPTOP-PK65VJLN MINGW64 ~
$ ls
-1.14-windows.xml
AppData/
Application Data@
Manvendra-GLA-CourseRegistration/
Manvendra-GLA-HealthSystem/
Manvendra-GLA-LibrarySystem/
Contacts/
Cookies/
CrossDevice/
Desktop/
Documents/
Downloads/
Favorites/
IdeaProjects/
IdeaSnapshots/
IntelliJ IDEA 2025.2.1/
Links/
Local Settings@
Music/
My Documents@
Myproject/
NTUSER.DAT
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TM.blf
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TM.blf.cnpf
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TMContainer00000000000000000001.regtrans-ms
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TMContainer00000000000000000001.regtrans-ms.cnpf
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TMContainer00000000000000000002.regtrans-ms
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaal}.TMContainer00000000000000000002.regtrans-ms.cnpf
NetHood/
OneDrive/
PrintHood@
Recent@
Saved Games/
Searches/
SendTo@
Start Menu @
TASK
Templates@
Videos/
ashu/
manvendra/
manvendra-GLA-CourseRegistration/
main.py
ntuser.dat.LOG1
ntuser.dat.LOG2
ntuser.ini

Manvendra@LAPTOP-PK65VJLN MINGW64 ~
$ cat library_info.txt
```

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Q5 — Question

Task 5 – Submission: Submit screenshots of important commands, the project directory, Git history, branch information, merge demonstration, and conflict and resolution demonstration. Students must be able to explain the commands used.

```
Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ cat library_info.txt
Name: Manvendra Singh Tomar
Roll Number: 241500188
Subject: Application Build and Release Automation Lab
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Semester: 5th

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Project Description:
The system will help manage library books and provide basic book management functionality.

Project Name: GLA Digital Library System

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ cd Ayush-GLA-LibrarySystem
bash: cd: Ayush-GLA-LibrarySystem: No such file or directory

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ cd Manvendra-GLA-LibrarySystem
bash: Manvendra-GLA-LibrarySystem: command not found

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ cat library_info.txt
bash: cat: library_info.txt: No such file or directory

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ cat library_info.txt
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Project Description:
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Project Name: GLA Digital Library System

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git branch
  book-feature
  library-update
* master

Manvendra@LAPTOP-PK65VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git log --oneline --graph --all
```

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Q5 — Question

Task 5 – Submission: Submit screenshots of important commands, the project directory, Git history, branch information, merge demonstration, and conflict and resolution demonstration. Students must be able to explain the commands used.

```
Project Name: GLA Digital Library System

Manvendra@LAPTOP-PK655VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git branch
  book-feature
  library-update
* master

Manvendra@LAPTOP-PK655VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ git log --oneline --graph --all
  c215eec (HEAD -> master) Finalize conflict resolution
  b02a695 Clean final merge resolution
  ac43de8 Resolve merge conflict
| \
| * 4c4646a (library-update) Update project name in library
| | 9704312 Modify project name in master
| /
> 7a2c8dc (book-feature) Add Book Management Module
* 5b2e714 Add project objective and description
  22551a7 Initial project setup

Manvendra@LAPTOP-PK655VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$ cat src/book.txt
Book Management Module

Description:
The Book Management Module is used to manage books in the online library system.

Features:
1. Add new books
2. Search books
3. Update book information
4. Delete books

Manvendra@LAPTOP-PK655VJLN MINGW64 ~/Manvendra-GLA-LibrarySystem (master)
$
```